MSE 370
Electrical, Optical, And Magnetic Properties of Materials
Fall Semester, 2025

Course description

In MSE 370, we will explore the fundamental theory behind the electrical, optical, and magnetic properties of materials, including semiconductors, metals, and insulators. Starting with a discussion of crystal structures and chemical bonding, we will study how and why certain materials show the properties that they do, and how electronic devices are built to take advantage of these properties. Using atomistic and electronic level descriptions, this course will provide a basic understanding of semiconductors, the origins of their properties, and a starting point for understanding device performance. The applications that will be covered include diodes, transistors, photodetectors, solar cells (photovoltaics), displays, light emitting diodes, lasers, optical fibers and optical communications, photonic devices, magnetic data storage, motors, transformers and spintronics. In-class lectures, regular quizzes, homework assignments and exams, and critical reading of published literature, will be utilized in this course. Lecture slides and reading material as well as announcements will be periodically uploaded to the course page on Brightspace.

Topics and Course Schedule

- Introduction: Overview of Electrical, Optical, and Magnetic Properties (Week 1)

- Part 1: Structure and Bonding in Solid-State Materials (Weeks 1 to 3)
  - Crystal Structures & Reciprocal Lattice
  - Chemical Bonding & Crystal Binding
  - Defects in Solids

- Part 2: Electrical Properties of Materials (Weeks 4 to 8)
  - Spectroscopic methods and energy levels in solids
  - Electronic structure of materials
  - Electrical properties of semiconductors
  - Carrier action and devices: pn Junctions, Transistors, etc.

- Part 3: Optical Properties of Materials (Weeks 9 to 12)
  - Overview of Optical Properties
  - Light Emitting Diodes (LEDs)
  - Photovoltaics, Solar Cell Materials
  - Other applications: Lasers, LCDs, Fibers

- Part 4: Magnetic Properties of Materials (Weeks 13 to 16)
  - Overview of Magnetic Properties
  - Types of Magnetic Materials
  - Applications: Motors, Data Storage, Superconductivity

Textbook

There is no single textbook that covers all the topics that will be taught in this course. Lecture notes/slides, homework assignments, and announcements will be available on Brightspace: *https://purdue.brightspace.com/d2l/login*. Additional reading material will include selected chapters of books and recent literature.

Chapters and material from the following textbooks will be utilized for this course:

- *Semiconductor Device Fundamentals, Pierret, 2nd edition.*
- *Materials Science and Engineering: An Introduction, Callister and Rethwisch*
- *Electronic materials and devices, Kasap, McGraw Hill 4th edition*
- *Electronic Properties of Materials, Rolf E. Hummel, Springer-Verlag, 1985*
- *Fundamentals of Ceramics, M. Barsoum, IoP, 1997*
- *The Electronic Structure and Chemistry of Solids, P.A. Cox, Oxford University Press, 1987*
- *Introduction to Solid State Physics, Charles Kittel, Wiley, 2004*
- *Solid State Materials Chemistry, Patrick M. Woodward, Cambridge University Press, 2021*

Course organization

The class will meet for 3 lectures per week, each lecture being 50 minutes long. Homework assignments will be provided periodically (typically every 1-2 weeks) and must be submitted by due date. Short quizzes will be given in class or on Brightspace to evaluate student progress. There will be no fixed schedule for quizzes and they will not be announced in advance. Two exams will be held in-class, roughly around the middle of October and end of November. There will be no final exam as such, but there will be a final "take home HW assignment". Optional office hours hosted by the instructor will be scheduled as and when required. Students are encouraged to reach out to the instructor via email any time.

Learning Outcomes

- Students understand essential theoretical concepts of crystal structures, chemical bonding, and point defects in solids.
- Students understand how the electronic structure of a material is characterized theoretically and experimentally.
- Students learn how to calculate electrical and optical properties from the electronic structure of materials.
- Students learn about the electrical properties of semiconductors, how they are affected by the presence of defects and dopants, and their relation to various applications.
- Students learn about how the concentrations and mobilities of electrons and holes affect the performance of various devices such as p-n junctions, transistors, solar cells, and light emitting diodes.
- Students understand how to characterize optical absorption spectra of materials and relate them to device performance, especially solar energy and light emission.

- Students learn how to calculate power conversion and other efficiencies of various optical and electrical devices.
- Students understand the fundamental phenomena giving rise to magnetism in materials.
- Students learn the differences between hard and soft magnets and how they are used in various applications.
- Students learn how to calculate magnetic moments, susceptibility, and related properties.
- Students will become comfortable with using electronic configurations, electronegativities, and bonding information to understand the electronic and magnetic properties of materials.

Grading Policy

Quizzes: **15%**
Homework Assignments: **40%** (5 HWs, 8% each)
Exam 1: **15%**
Exam 2: **15%**
Final report/assignment: **15%**

Letter grades (A+, A, B+, B, …) will be assigned based on the total score out of 100. Exact point ranges will be determined later based on the performance of the class.

Prerequisites

Students should have working knowledge of college-level physics, chemistry, mathematics, and materials science, as well as access to a working computer. Familiarity with materials structure, chemical bonding, and differential equations will also be helpful in completing this course. Familiarity with nanoHUB may be required to complete some homework assignments.